Regional Income Growth and House Price Dynamics in the Netherlands

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Set-up your environment

```r
require(tidyverse)
```

```
## Loading required package: tidyverse
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
require(rmarkdown)
```

```
## Loading required package: rmarkdown
```

```r
require(yaml)
```

```
## Loading required package: yaml
```

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Tutorial group 3.3

C. Schouwenaar

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

The Dutch housing market has been under pressure for several years now. Between 2013 and 2023 the average price of houses has more than doubled, while the income growth in the same period considerably lagged behind. The increasing gap between housing costs and purchasing power is not only an economic phenomenon, but also a social problem; it determines who does have access to a stable living situation and who doesn't.

It's striking that this development differs significantly across regions in the Netherlands. In the Randstad, prices increased much more than in the peripheral regions like Groningen or Zeeland, while the income development per region differs significantly too. Factors such as access to big employers, education level and local economic structure play an important role in this shift. And it is because of this regional variety that it's interesting to investigate the relation between income and house prices.

This project investigates the relationship between regional income growth and house price dynamics in Dutch provinces, based on CBS-data and the programming tool R. The goal is to discover in which regions the housing market detached most from the local income development.

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

hoge_veerkracht_2024 <- #read.csv("C:/Users/vince/Downloads/hoge-veerkracht-2024.csv", header=FALSE, sep=",")

write.csv(hoge_veerkracht_2024, "data/veerkracht2024.csv")

Inkomen_van_huishoudens___huishoudenskenmerken___regio___indeling_2024___02062026_164030 <- #read.csv("C:/Users/vince/Downloads/Inkomen_van_huishoudens___huishoudenskenmerken___regio___indeling_2024___(header=FALSE, sep=",")

write.csv(Inkomen_van_huishoudens___huishoudenskenmerken___regio___indeling_2024___02062026_164030, "data/inkomenperregio.csv")

«««< HEAD Bestaande_koopwoningen <- #read.csv("Bestaande_koopwoningen__gemiddelde_verkoopprijzen_regio_02062 header = FALSE, sep=",")

write.csv(bestaande_koopwoningen_gemiddelde_verkoopprijzen_regio_020 "data/bestaande koopwoningen.csv")

«««< HEAD

```r
#Bestaande_koopwoningen <- #read.csv("Bestaande_koopwoningen_gemiddelde_verkoopprijzen_regio_02062026_1

#write.csv(bestaande_koopwoningen_gemiddelde_verkoopprijzen_regio_02062026_164630, "data/bestaande koop

#data_2015 <- read.csv("C:/Uni/Year 1/Period 6/Programming for Economics/2015.csv", header=FALSE, sep="
X2015 <- read.csv("data/data2015.csv")
write.csv(X2015, "data/data2015.csv", row.names = FALSE)

#data_2016 <- read.csv("C:/Uni/Year 1/Period 6/Programming for Economics/2016.csv", header=FALSE, sep="
X2016 <- read.csv("data/data2016.csv")
write.csv(X2016, "data/data2016.csv", row.names = FALSE)

#data_2017 <- read.csv("C:/Uni/Year 1/Period 6/Programming for Economics/2017.csv", header=FALSE, sep="
X2017 <- read.csv("data/data2017.csv")
write.csv(X2017, "data/data2017.csv", row.names = FALSE)

#data_2018 <- read.csv("C:/Uni/Year 1/Period 6/Programming for Economics/2018.csv", header=FALSE, sep="
X2018 <- read.csv("data/data2018.csv")
write.csv(X2018, "data/data2018.csv", row.names = FALSE)

#data_2019 <- read.csv("C:/Uni/Year 1/Period 6/Programming for Economics/2019.csv", header=FALSE, sep="
X2019 <- read.csv("data/data2019.csv")
write.csv(X2019, "data/data2019.csv", row.names = FALSE)

#data_2020 <- read.csv("C:/Uni/Year 1/Period 6/Programming for Economics/2020.csv", header=FALSE, sep="
 X2020 <- read.csv("data/data2020.csv")
write.csv(X2020, "data/data2020.csv", row.names = FALSE)

#data_2021 <- read.csv("C:/Uni/Year 1/Period 6/Programming for Economics/2021.csv", header=FALSE, sep="
X2021 <- read.csv("data/data2021.csv")
write.csv(X2021, "data/data2021.csv", row.names = FALSE)

#data_2022 <- read.csv("C:/Uni/Year 1/Period 6/Programming for Economics/2022.csv", header=FALSE, sep="
X2022 <- read.csv("data/data2022.csv")
write.csv(X2022, "data/data2022.csv", row.names = FALSE)

#data_2023 <- read.csv("C:/Uni/Year 1/Period 6/Programming for Economics/2023.csv", header=FALSE, sep="
X2023 <- read.csv("data/data2023.csv")
write.csv(X2023, "data/data2023.csv", row.names = FALSE)
```

```r
# prijswoningen <- read.csv(C:\Uni\Year 1\Period 6\Programming for Economics/gem_prijs_woning_prov)
# Gem_prijs_woning_prov <- read.csv("goede data/woningprijsprov.csv")
# write.csv(Gem_prijs_woning_prov, "goede data/woningprijsprov.csv")
```

joining

```r
data2015 <- read.csv("data/data2015.csv")
data2015$jaar <- 2015
data2016 <- read.csv("data/data2016.csv")
data2016$jaar <- 2016
```

```r
data2015$X.5 <- NULL
data2016$X.5 <- NULL

data2015$X.7 <- NULL
data2016$X.7 <- NULL

data1516 <- rbind(data2015, data2016)

data2017 <- read.csv("data/data2017.csv")
data2017$jaar <- 2017
data2018 <- read.csv("data/data2018.csv")
data2018$jaar <- 2018

data2019 <- read.csv("data/data2019.csv")
data2019$jaar <- 2019

data2020 <- read.csv("data/data2020.csv")
data2020$jaar <- 2020
data2021 <- read.csv("data/data2021.csv")
data2021$jaar <- 2021
data2022 <- read.csv("data/data2022.csv")
data2022$jaar <- 2022
data2023 <- read.csv("data/data2023.csv")
data2023$jaar <- 2023

data2017$X.5 <- NULL
data2018$X.5 <- NULL
data2019$X.5 <- NULL
data2020$X.5 <- NULL
data2021$X.5 <- NULL
data2022$X.5 <- NULL
data2023$X.5 <- NULL

data1718 <- rbind(data2017, data2018)

Data15161718 <- rbind(data1516, data1718)

write.csv(Data15161718, "data/data15161718.csv")

datacombined15tot23 <- rbind(Data15161718, data2019, data2020, data2021, data2022, data2023)

write.csv(datacombined15tot23, "data/datacombined15tot23.csv")

datacombined15tot23 <- datacombined15tot23[datacombined15tot23$X != 13, ]

colnames(datacombined15tot23)
```

```
##  [1] "X.6"             "X.4"             "X.3"             "X.2"
##  [5] "X.1"             "X"               "Regio.s"         "x.1.000"
##  [9] "X."              "X1.000.euro...4" "X1.000.euro...5" "X1.000.euro...6"
## [13] "X1.000.euro...7" "jaar"
```

```
saveRDS(datacombined15tot23, "goede data/datacombi1523.rds")
```

2.2 Provide a short summary of the dataset(s)

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

Part 3 - Quantifying

3.1 Data cleaning

Data cleaning codering datacombined15tot23goed.

```
datacombined15tot23goed <- read.csv("goede data/datacombined15tot23goed.csv")
names(datacombined15tot23goed)[names(datacombined15tot23goed) == "X1.000.euro...6"] <- "gem_st_inkomen_

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "X1.000.euro...7"] <- "med_st_inkomen_

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "X1.000.euro...8"] <- "gem_bstb_inkomen

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "X1.000.euro...9"] <- "med_bstb_inkomen

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "X1.000.euro...8"] <- "gem_bstb_inkomen

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "X1.000.euro...9"] <- "med_bstb_inkomen

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "X1.000.euro...9"] <- "med_bstb_inkomen

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "Regio.s"] <- "Provincies"

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "`nr_huishoudens"] <- "ant_hh_p_provin

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "X"] <- "Provincie_nr"

names(datacombined15tot23goed)[names(datacombined15tot23goed) == "x.1.000"] <- "Nr_huishoudens"

write.csv(datacombined15tot23goed, "goede data/datacombined15tot23goed.csv", row.names = FALSE)

saveRDS(datacombined15tot23goed, "goede data/datacombined15tot23goed.rds")
```

```
datacombined15tot23goed <- datacombined15tot23goed[datacombined15tot23goed$Provincies != "Bron: CBS", ]

datacombined15tot23goed$X. <- NULL
datacombined15tot23goed$...1 <- NULL
```

Data cleaning codering woningprijsprov

```
woningprijsprov <- read.csv("goede data/goedehuizenprijsenperprov.csv")
woningprijsprov$`1995` <- format(woningprijsprov$`1995`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2000` <- format(woningprijsprov$`2000`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2005` <- format(woningprijsprov$`2005`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2010` <- format(woningprijsprov$`2010`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2015` <- format(woningprijsprov$`2015`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2020` <- format(woningprijsprov$`2020`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2021` <- format(woningprijsprov$`2021`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2022` <- format(woningprijsprov$`2022`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2023` <- format(woningprijsprov$`2023`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2024` <- format(woningprijsprov$`2024`, big.mark = ".", scientific = FALSE)
woningprijsprov$`2025` <- format(woningprijsprov$`2025`, big.mark = ".", scientific = FALSE)
woningprijsprov$...1 <- NULL

saveRDS(woningprijsprov, "goede data/gem_huizenprijs_per_provincie.RDS")
```

```
gem_huizenprijs_per_provincie <- readRDS("goede data/gem_huizenprijs_per_provincie.RDS")
write.csv(gem_huizenprijs_per_provincie, "goede data/goedehuizenprijsenperprov.csv", row.names = FALSE)
```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1 maken van nieuwe variabele 1, we hebben hier het gemiddelde besteedbare inkomen van iedere provincie over 8 jaar. aldus het gemiddeld

```
gem_inkomen_provincie <- aggregate(gem_bstb_inkomen_x_100 ~ Provincies,
                                   data = datacombined15tot23goed,
                                   FUN = mean)

write.csv(gem_inkomen_provincie, "goede data/gem_inkomen_provincie.csv", row.names = FALSE)

colnames(gem_inkomen_provincie)[colnames(gem_inkomen_provincie) == "gem_bstb_inkomen_x_100"] <- "gem_bs

write.csv(gem_inkomen_provincie, "goede data/gem_inkomen_provincie.csv", row.names = FALSE)
```

Variable 2 maken van nieuwe variabele 2 gebruik gemaakt van de oorspronkelijke dataset:datacombined15tot23goed. waarin we de gewogen gemiddeldes van het inkomen van verschillende provincies: binnen en buiten de randstad hebben gerepresenteerd

de vorm van de huidige gecleande dataset aanpassen om dan zo verder te kunnen werken met een vernieuwde dataset en ook een vernieuwde dataset. omdat we voor deze nieuwe variabele eigenlijk onze twee datasets moeten samenvoegen.

```r
woningprijsprov <- read.csv("goede data/goedehuizenprijsenperprov.csv", check.names = FALSE)

woningprijsprov <- woningprijsprov[!grepl("Bron", woningprijsprov$`Regio.s`), ]

woningprijsprov$Onderwerp <- NULL
woningprijsprov$`...4` <- NULL

woningprijsprov_lang <- reshape(woningprijsprov,
                                varying = c("2015", "2020", "2021", "2022", "2023"),
                                v.names = "gem_huizenprijs",
                                timevar = "jaar",
                                times = c(2015, 2020, 2021, 2022, 2023),
                                direction = "long")

woningprijsprov_lang$id <- NULL

write.csv(woningprijsprov_lang, "goede data/goedehuizenprijsenperprov_lang.csv", row.names = FALSE)
```

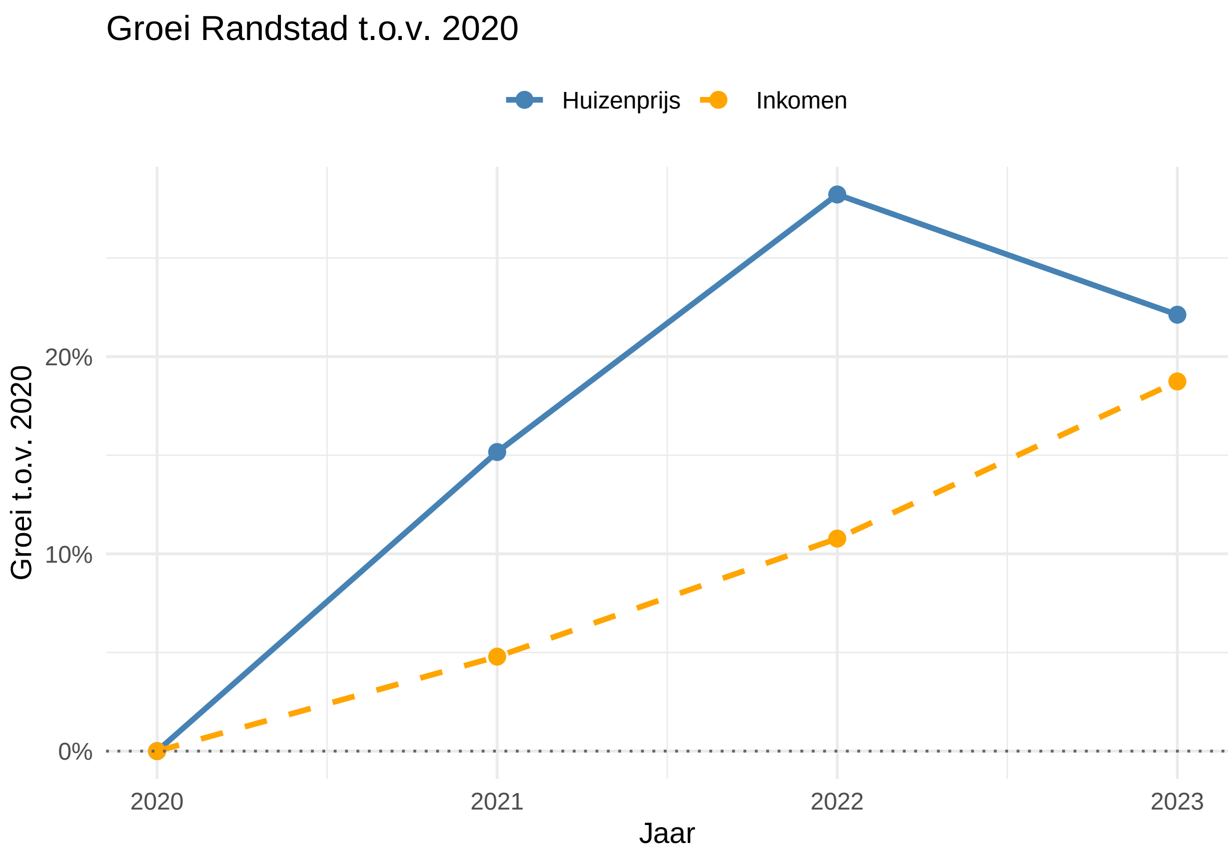
3.3 Visualize temporal variation

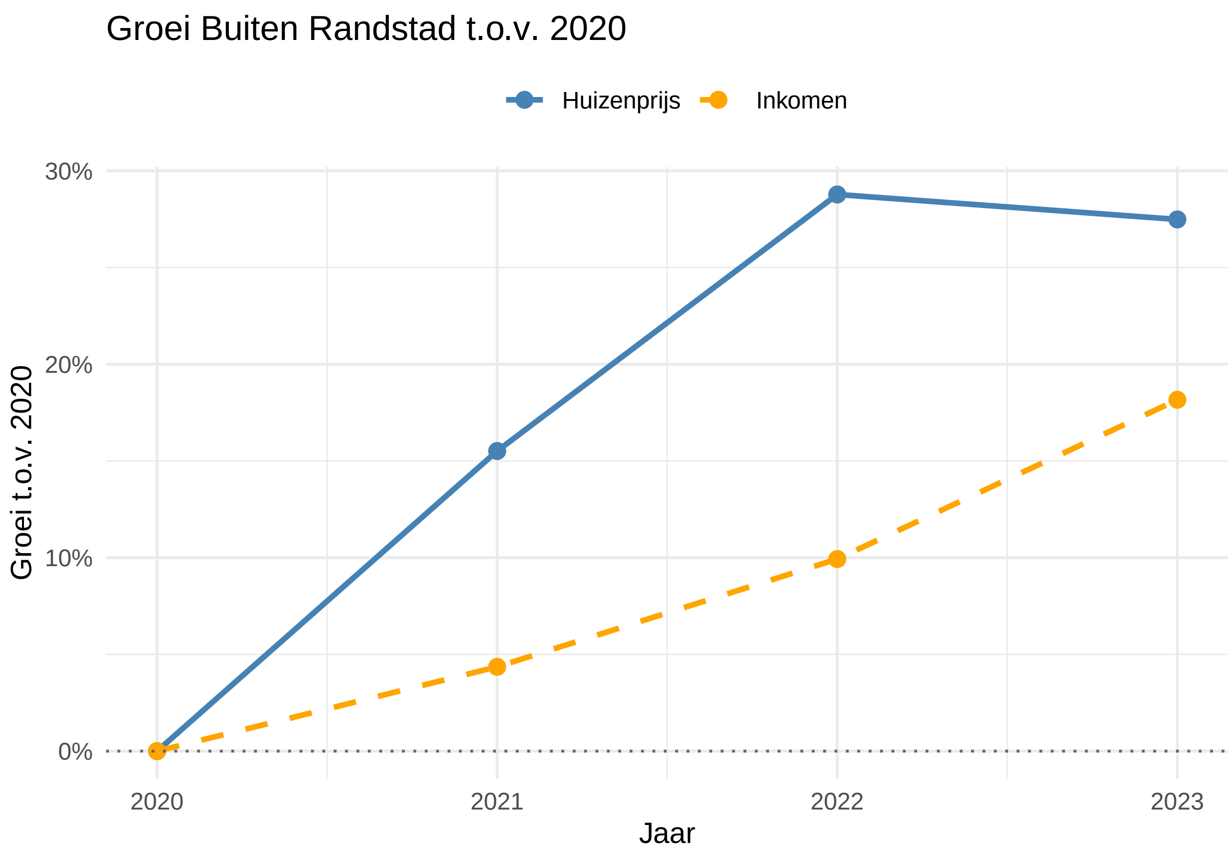
```r
library(tidyverse)
library(scales)
```

```
##
## Attaching package: 'scales'

## The following object is masked from 'package:purrr':
##
##     discard

## The following object is masked from 'package:readr':
##
##     col_factor
```

```r
# Data inladen
df <- read.csv("goede data/randstad_vs_buiten_randstad.csv", stringsAsFactors = FALSE)

# Cumulatieve groei berekenen
df_plot <- df %>%
  arrange(regio, jaar) %>%
  group_by(regio) %>%
  mutate(
    groei_huizen = (gem_huizenprijs / first(gem_huizenprijs)) - 1,
    groei_inkomen = (gem_bstb_inkomen_euro / first(gem_bstb_inkomen_euro)) - 1
  ) %>%
  ungroup()

# Data omzetten naar lang formaat
df_long <- df_plot %>%
  pivot_longer(cols = c(groei_huizen, groei_inkomen),
```

```r
               names_to = "type",
               values_to = "groei") %>%
  mutate(type = recode(type,
                       "groei_huizen" = "Huizenprijs",
                       "groei_inkomen" = "Inkomen"))

# Functie voor grafiek
maak_grafiek <- function(data, titel) {
  ggplot(data, aes(x = jaar, y = groei, color = type, linetype = type)) +
    geom_line(linewidth = 1) +
    geom_point(size = 2.5) +
    geom_hline(yintercept = 0, linetype = "dotted", color = "grey40") +
    scale_y_continuous(labels = percent_format(accuracy = 1)) +
    scale_x_continuous(breaks = c(2020, 2021, 2022, 2023)) +
    scale_color_manual(values = c("Huizenprijs" = "steelblue", "Inkomen" = "orange")) +
    scale_linetype_manual(values = c("Huizenprijs" = "solid", "Inkomen" = "dashed")) +
    labs(title = titel, x = "Jaar", y = "Groei t.o.v. 2020",
         color = "", linetype = "") +
    theme_minimal() +
    theme(legend.position = "top")
}

# Automatisch beide grafieken tonen
grafieken_tonen <- function() {
  print(maak_grafiek(
    df_long %>% filter(regio == "Randstad (incl. Flevoland)"),
    "Groei Randstad t.o.v. 2020"
  ))
  print(maak_grafiek(
    df_long %>% filter(regio == "Buiten Randstad"),
    "Groei Buiten Randstad t.o.v. 2020"
  ))
}

grafieken_tonen()
```


Groei Randstad t.o.v. 2020
Huizenprijs
Inkomen
Groei t.o.v. 2020
20%
10%
0%
2020
2021
2022
2023
Jaar



3.4 Visualize spatial variation

```r
install.packages(c("sf", "ggplot2"))
```

```
## # Downloading packages -------------------------------------------------------
## - Downloading sf from CRAN ...                  OK [22.5 Mb in 2.6s]
## - Downloading classInt from CRAN ...            OK [486.3 Kb in 0.63s]
## - Downloading e1071 from CRAN ...               OK [576.4 Kb in 0.71s]
## - Downloading proxy from CRAN ...               OK [168 Kb in 0.62s]
## - Downloading s2 from CRAN ...                  OK [4.1 Mb in 0.97s]
## - Downloading Rcpp from CRAN ...                OK [2.2 Mb in 0.94s]
## - Downloading wk from CRAN ...                  OK [1.6 Mb in 0.73s]
## - Downloading units from CRAN ...               OK [582.6 Kb in 0.69s]
## Successfully downloaded 8 packages in 9 seconds.
##
## The following package(s) will be installed:
## - classInt [0.4-11]
## - e1071    [1.7-17]
## - ggplot2  [4.0.3]
## - proxy    [0.4-29]
## - Rcpp     [1.1.1-1.1]
## - s2       [1.1.11]
## - sf       [1.1-1]
```

```
## - units    [1.0-1]
## - wk       [0.9.5]
## These packages will be installed into "~/Documents/GitHub/R-project-group-3-3/renv/library/macos/R-4
##
## # Installing packages --------------------------------------------------------
## - Installing proxy ...                          OK [installed binary and cached]
## - Installing e1071 ...                          OK [installed binary and cached]
## - Installing classInt ...                       OK [installed binary and cached]
## - Installing Rcpp ...                           OK [installed binary and cached in 0.34s]
## - Installing wk ...                             OK [installed binary and cached]
## - Installing s2 ...                             OK [installed binary and cached]
## - Installing units ...                          OK [installed binary and cached]
## - Installing sf ...                             OK [installed binary and cached in 0.39s]
## - Installing ggplot2 ...                        OK [linked from cache]
## Successfully installed 9 packages in 1.3 seconds.
```

```r
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.5.3, PROJ 9.5.1; sf_use_s2() is TRUE
```

```r
library(ggplot2)
nl_map <- st_read("https://cartomap.github.io/nl/wgs84/provincie_2024.geojson")
```

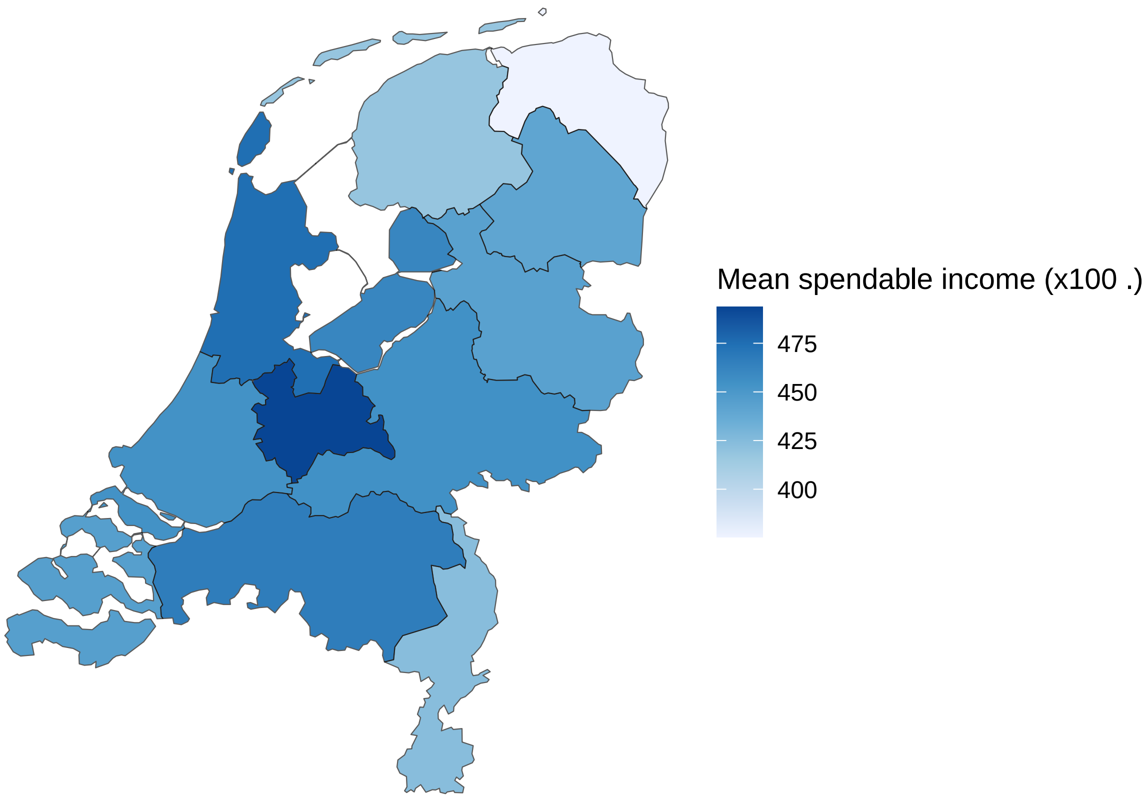
```
## Reading layer ‘provincie_2024’ from data source
##   ‘https://cartomap.github.io/nl/wgs84/provincie_2024.geojson’
##   using driver ‘GeoJSON’
## Simple feature collection with 12 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 3.358 ymin: 50.751 xmax: 7.218 ymax: 53.554
## Geodetic CRS:  WGS 84
```

```r
nl_map$statnaam
```

```
##  [1] "Groningen"     "Fryslân"       "Drenthe"       "Overijssel"
##  [5] "Flevoland"     "Gelderland"    "Utrecht"       "Noord-Holland"
##  [9] "Zuid-Holland"  "Zeeland"       "Noord-Brabant" "Limburg"
```

```r
my_data <- data.frame(
  Provincies = c("Groningen", "Fryslân", "Limburg", "Drenthe", "Overijssel",
                 "Zeeland", "Zuid-Holland", "Gelderland", "Flevoland",
                 "Noord-Brabant", "Noord-Holland", "Utrecht"),
  mean_income = c(375.5556, 418.3333, 423.7778, 440.7778, 443.6667,
                  445.1111, 453.8889, 454.7778, 461.2222, 466.7778,
                  474.8889, 493.5556)
)
nl_map <- merge(nl_map, my_data, by.x = "statnaam", by.y = "Provincies")
ggplot(nl_map) +
  geom_sf(aes(fill = mean_income)) +
  scale_fill_distiller(
    palette = "Blues",
```

```
    direction = 1,         # 1 = darker means higher value
    name = "Mean spendable income (x100 €)"

  ) +
  theme_void()
```




```
  labs(
    title = "Mean Spendable Income per Province",
    subtitle = "Average over 2015-2023"
  )
```

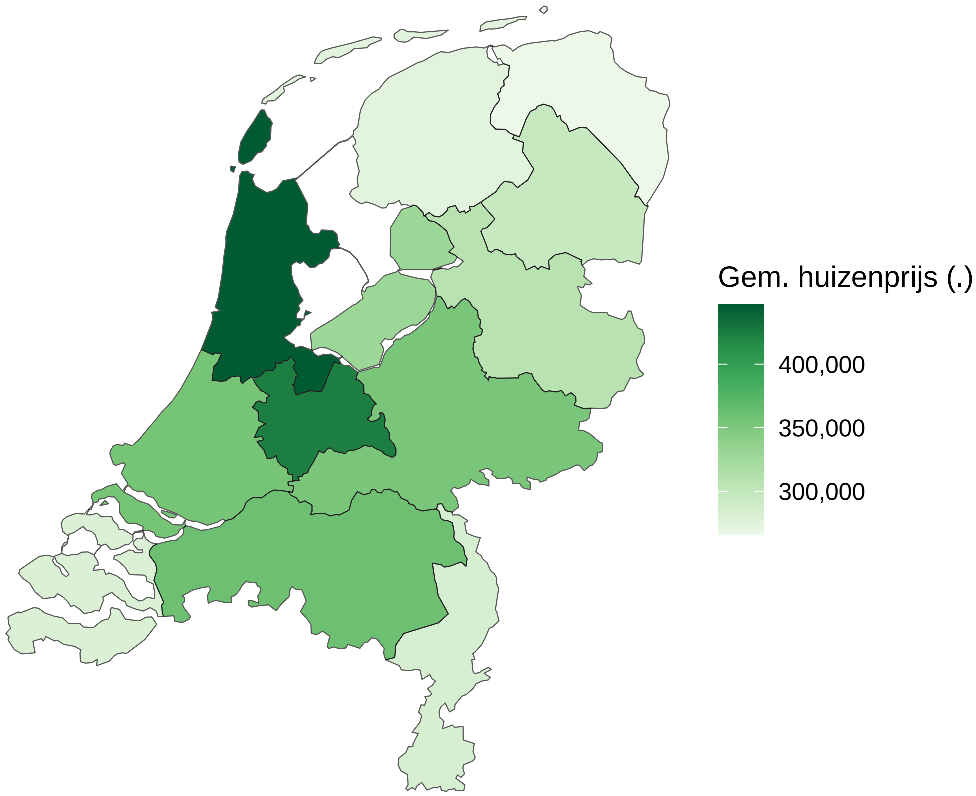
```
## <ggplot2::labels> List of 2
##  $ title   : chr "Mean Spendable Income per Province"
##  $ subtitle: chr "Average over 2015-2023"
```

```
  library(sf)
library(ggplot2)

# Stap 1: Kaart laden
nl_map <- st_read("https://cartomap.github.io/nl/wgs84/provincie_2024.geojson")
```

```
## Reading layer `provincie_2024' from data source
##   `https://cartomap.github.io/nl/wgs84/provincie_2024.geojson'
##   using driver `GeoJSON'
```

```
## Simple feature collection with 12 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 3.358 ymin: 50.751 xmax: 7.218 ymax: 53.554
## Geodetic CRS:  WGS 84
```

```r
# Stap 2: Jouw data
my_data2 <- data.frame(
  Provincies = c("Groningen", "Fryslân", "Drenthe", "Overijssel", "Flevoland",
                 "Gelderland", "Utrecht", "Noord-Holland", "Zuid-Holland",
                 "Zeeland", "Noord-Brabant", "Limburg"),
  huizenprijs = c(265840.0, 274763.0, 296620.4, 307718.0, 329630.8,
                  353163.8, 423367.6, 446549.2, 355059.0,
                  280299.6, 360750.2, 283216.8)
)

# Stap 3: Samenvoegen
nl_map <- merge(nl_map, my_data2, by.x = "statnaam", by.y = "Provincies")

# Stap 4: Kaart plotten
ggplot(nl_map) +
  geom_sf(aes(fill = huizenprijs)) +
  scale_fill_distiller(
    palette = "Greens",
    direction = 1,
    name = "Gem. huizenprijs (€)",
    labels = scales::comma
  ) +
  theme_void()
```




```
  labs(
    title = "Gemiddelde Huizenprijs per Provincie",
    subtitle = "Gemiddelde over de laatste jaren"
  )
```

```
## <ggplot2::labels> List of 2
##  $ title   : chr "Gemiddelde Huizenprijs per Provincie"
##  $ subtitle: chr "Gemiddelde over de laatste jaren"
```

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.